Brazil in a Global Atlas of Lyrical Themes: Chart Repertoires and Social Context

Data freeze: 11 October 2026

Abstract

We study how lyrical themes circulate across music markets and how they relate to chart exposure and social conditions. From an archive of 47.43 million source records, the final analytical corpus includes 153,501 classified recording IDs in 68 markets. Every recording in this final corpus has a classification; recordings without one are excluded and documented. Between 2017 and 2021, status and conflict themes increase in a fixed 55-market panel, with different regional patterns. Genre and release-group decompositions show that changes in repertoire composition account for part of these trends. Four exploratory theme clusters broadly separate Latin American and Asian repertoires. In 51 Top-200 markets, romantic and sexually explicit songs both have longer observed chart presence than songs without those themes. National Gini and homicide indicators show descriptive associations, without establishing predictive accuracy or causation. Human validation metrics remain pending because reviewer-level labels are unavailable. Figures and a market atlas distinguish song counts, chart days and recorded plays.

1. Research question

How do sexual, romantic, gender, material, and crime and conflict themes differ across music markets, and how do they relate to chart dynamics and external indicators? Brazil is a focal case, compared with other markets using the same measures. The analysis separates sexual content from objectification and misogyny, and thematic presence from endorsement. This allows several aspects of a song to coexist without collapsing them into one moral score.

The study combines historical charts, lyrics, structured Gemini classification and social indicators matched by year. A specialist check was reported by the project lead; an independent, measured validation is still needed. We examine the songs circulating in each market, changes over time and associations with social conditions. A market identifies where a recording appears in the charts, not the artist's nationality or the listener's identity.

Six axes organize the analysis: **intimacy** (romance, heartbreak, reciprocity and sexual relations); **gender and power** (agency, objectification and misogyny); **status and inequality** (money, materialism and wealth); **violence and security** (crime, weapons and drugs); **digital environment** (access, search and connectivity); and **cultural circulation** (identity, genres and chart persistence). Brazil is the focal case. The atlas lets every market inspect its own profile, position and coverage.

2. Related research and contribution

Research on lyrics has already documented long-term changes in popular music. Brand, Acerbi and Mesoudi studied emotional expression over five decades of song lyrics, showing how popular songs can be used to examine cultural transmission and emotional change [9]. Their emotional measures do not by themselves distinguish romance, sexual content, wealth, weapons or the stance taken toward violence. Our study makes those distinctions explicit and compares their circulation across markets.

Varnum and colleagues analyzed 14,661 songs in US charts between 1958 and 2016. They linked simpler lyrics to a larger supply of new songs and to chart success [10]. Parada-Cabaleiro and colleagues examined changes in lyrical complexity, repetition and emotional language across genres [11]. Together, these studies show why an aggregate change in lyrics needs to be examined alongside the supply and type of songs entering the charts. We address that issue through genre and release-group decompositions, while recognizing that broad artist-derived tags and incomplete dates cannot fully explain repertoire selection.

International listening also varies with cultural and social context. Liu, Hu and Schedl studied cross-country music preferences in relation to culture, socioeconomic conditions and friendship [12]. Schedl and colleagues identified country archetypes from listening patterns and used them in music recommendation [13]. Those studies motivate regional comparisons, but listening-pattern clusters and clusters of lyrical themes measure different things. A country can share themes with another without sharing the same songs or listeners.

The added contribution is a common 26-dimension lyrical taxonomy linked to recording IDs, circulation markets, years, chart days, recorded streams and dated social indicators. It separates romantic affection, sexual explicitness, gender representation and material or violent themes instead of combining them into one score. It also compares counting songs with counting chart exposure, and tests how much temporal change is associated with genre and release-group composition. This is a descriptive extension of earlier work, not evidence that chart lyrics directly measure national values. Its stronger multilingual claims require the independent human validation described below.

3. Corpus and method

3.1 Sources, units, and coverage

Table 1 inventories the unified source catalog [1]. MGD and Kaggle preserve overlapping Spotify charts; 19,942,141 exact native-ID equivalences were identified in the earlier overlap audit. Their row counts are therefore not additive evidence of independent events. We normalize Spotify recording IDs and count each ID once per market for the all-history comparison, and once per market–year for the annual comparison. Top 200 and Viral 50 are combined to describe the broader chart reper-

Table 1: Source archive before analytical deduplication. Counts include source territories; they are not audience sizes.

Source / chart	Records	Period
MGD / Spotify Top 200	21,257,472	2017–22
Kaggle / Spotify Top 200	20,321,904	2017–21
Kaggle / Spotify Viral 50	5,851,610	2017–21
YouTube / popular videos	3,037	Sep. 2026
Pro-Música / aggregate	50	Aug. 2026
Total stored source records	47,434,073	

toire; source-specific results remain available separately.

The geographical analysis excludes GLOBAL, Hong Kong and Puerto Rico under the established market policy, and keeps Taiwan as a separate market. The initial chart inventory contains 222,102 distinct recording IDs and 652,121 recording–market pairs in 68 markets. We exclude 68,601 recording IDs without a usable classification, corresponding to 190,674 recording–market pairs. The final analytical corpus therefore contains 153,501 recording IDs and 461,447 recording–market pairs across 386 market–year cells. Classification is available for 100% of this final corpus. The selection retains 70.8% of the initial pairs, ranging from 59.7% to 79.1% by market; the remaining 29.2% are outside the analytical corpus. These selection rates stay visible in tables so that a complete final corpus is not confused with complete coverage of the original charts. Markets enter displayed comparisons with at least 300 classified recordings. This restriction defines a study of the classified repertoire and does not demonstrate that excluded songs have the same thematic profile.

The main window starts on 1 January 2017 and ends on 13 March 2022. The partial 2022 repertoire is described but excluded from annual contextual correlations and search models. The 2026 YouTube and Pro-Música captures are supplementary inventories, not extensions of the historical series. A YouTube video remains a video-level item, and the most-popular endpoint is not a historical YouTube Music Top Songs archive. Chartmetric pilot files and source capability records are retained as acquisition evidence, outside the counted source archive.

3.2 Classification and validation status

Lyrics are acquired through the configured provider cascade, with source, version, and SHA-256 text hash preserved. Gemini receives the supplied lyric text and returns schema-constrained JSON. The model is `gemini-3.8-flash`; the taxonomy and prompt are `lyrics-classification-v1` and `lyrics-classification-prompt-v2`. The contract contains 26 ordinal dimensions in six groups: sexuality, relationships, gender representation, material status, antisocial content, and territorial identity. It also codes consent, target gender, representation roles, and stance. Generation uses Vertex AI; batch outputs are reconciled into an append-only PostgreSQL history.

The project lead reported that specialists completed a check of classifications and lyric-to-recording links. The archived confirmation does not include the number of reviewed items, individual reviewer scores or error counts. It therefore cannot establish measured classification accuracy. An earlier review contained 58 candidates and 22 AI-assisted text inspections, without an independent

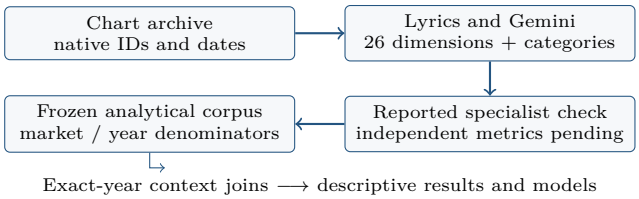


Figure 1: Research workflow. Source records and model histories remain traceable; the analytical joins use explicit recording and time keys.

human reference set; its blank review fields cannot support agreement or F1 estimates. The analytical freeze uses the latest result per document for the fixed model, taxonomy and prompt, then selects the latest applicable recording result. Text hashes, research authorization and existing exclusion decisions define the eligible version. No confidence cutoff is applied.

An independent validation package now contains 3,381 provisional candidate recordings and separate blind sheets for two reviewers. None of these new sheets has been scored. Sampling spans circulation market, an exclusive broad genre group and maximum thematic intensity from 0 to 3, excluding the earlier 58 review documents. Actual lyric language is unavailable in the frozen metadata: all language strata remain unresolved. The sample must be revised after independently confirming languages before it can be described as language-stratified. Both reviewers will score all retained items independently, using language-qualified reviewers where needed, followed by documented adjudication. The accompanying protocol defines counts, inclusion weights, agreement, binary precision/recall/F1 at intensity ≥ 2, ordinal confusion matrices, quadratic-weighted kappa, macro F1 and mean absolute error, with recording-level intervals and errors by language and theme. These metrics are currently unestimated, not zero. The present results depend on the frozen AI labels.

The freeze is 11 October 2026, 23:39 UTC (20:39 in São Paulo): 155,010 of 159,111 lyric documents are classified (97.4%), representing 154,979 recording identities before selection. The selected corpus has 154,995 documents, 154,964 recording identities, and 154,942 linked native IDs before geographical restrictions. Errors, instrumental, ambiguous and other unscored states remain separate from score zero; full counts accompany the analysis.

Scores run from 0 (absent) to 3 (central). A score of 2 or 3 means that a theme is clearly present. Each percentage is the share of classified recording IDs meeting this threshold. Each ID counts once per market and year; excluded recordings remain in the selection flow and are never treated as theme absence. The scores describe the text, not the artist's conduct. Separate fields record consent and whether antisocial content is condemned, neutral, ambivalent, normalized or glorified.

3.3 Chart dynamics and external context

Persistence counts days in the charts, including re-entries. Calendar span is reported separately. Top-50 renewal uses Jaccard distance between adjacent daily charts: how much the two sets of recordings differ. Comparisons across missing dates are excluded. Genre tags come from artist metadata and can overlap, so their shares need not

sum to 100%.

International trends use the same 55 markets in all five complete years, with at least 300 classified recording IDs and 50% classification coverage in every market–year. Each market has equal weight; interquartile ranges describe market dispersion, not confidence intervals. The source sensitivities use 52 balanced Top-200 markets, 51 MGD-only markets, and 55 Viral-50 markets; the chart-family comparison uses the same 52 markets in both families. Additional panels require 70% classification coverage throughout (8 markets), or pre-existing chart eligibility and a 1 January 2017 source start (46 markets). A distinct-recording yearly pool counts each ID once across the 55 markets. All denominators and membership lists are exported; partial 2022 is excluded.

A further analysis links the frozen classifications to 17,268,835 daily MGD chart entries in the same 51 markets during 2017–2021. Identical duplicate entries are counted once; source files are preserved. Each market–year is measured three ways: one vote per recording, weight by days in the charts, and weight by recorded chart streams. Each measure uses its own classified denominator and coverage. Genre profiles use the eight largest Brazilian artist-derived tags, selected by recording count before examining theme scores. Social indicators are also compared with all three weightings using the same complete-data markets within each comparison.

Context is joined by exact market and reference year: World Bank income, Gini, unemployment, population and internet indicators; UNDP Gender Inequality Index (GII); OECD SIGI for its 2019 edition; and UNODC homicide indicators [4, 5, 6, 7]. Missing values stay missing, without interpolation. SIGI is not converted into an annual time series. OONI connectivity tests are kept separately from music and search measures. The contextual export verifies unique keys, value ranges, and raw-response hashes.

Google Trends exports retain term, language, query batch, sampling period, and raw hash. The historical panel has 2,150 series–year summaries from 383 local term/language/batch series. Means are weighted by the days represented by each bucket; weeks crossing years are split. Values reported as < 1 contribute bounds $[0, 1)$ rather than an invented point estimate. The index is relative search interest, normalized to the query's place and period, not visits or counts of people [3].

3.4 Regional comparisons and repertoire composition

We compare 2017 with 2021 in the fixed 55-market panel. Regional summaries give each market equal weight and retain a fixed membership: Latin America (16), Nordic countries (5), East Asia (Japan and Taiwan only), Southeast Asia (5), six Anglophone markets and the 21 remaining panel markets. Membership lists are exported. These groups describe the available panel; they are not representative samples of entire world regions.

For each theme, a symmetric decomposition separates changes within repertoire groups from changes in the groups' shares. The within-group term weights the change in thematic prevalence by the group's average share in the two years. The composition term weights the change in group share by its average thematic prevalence. Their sum exactly reproduces the total change. We calculate this separately by broad genre and catalogue release group, within each market, then average markets. The two decompositions answer different questions and must not be added together.

Overlapping artist tags are assigned to one broad genre through a fixed priority rule published in the analysis script; untagged songs remain a group. Genre tags cover 55.8% of the 199,496 classified endpoint recording–market–year observations. Catalogue release dates cover 75.3%; dates later than the recording's first chart year are treated as unknown. Release groups are through 2016, 2017–2018, 2019–2020, 2021 onward and unknown. Catalogue dates can refer to editions or reissues rather than first artistic release. When a group appears in only one endpoint, its contribution is assigned to composition, with no invented within-group change. Actual lyric languages are absent, so no language decomposition is claimed.

Regional changes and decomposition terms have 95% percentile intervals from 400 recording-level Poisson bootstrap replicates, using seed 20261012. Each recording receives one weight shared across all its markets and years. This preserves the dependence caused by international hits. Intervals condition on the observed markets and frozen AI labels; they do not include unmeasured classification error or establish population-level cultural change.

3.5 Theme clusters and chart longevity

We cluster the 68 markets using all 26 theme percentages from the combined historical period, 2017 through 13 March 2022. We standardize each dimension across markets before Ward hierarchical clustering, compare two through eight clusters and choose the largest mean silhouette score, a measure of separation. This is an exploratory summary without external or resampling-based cluster validation.

Longevity uses the classified MGD Top-200 panel in 51 markets. We sum observed daily chart appearances from 2017 through 2021 for each recording–market pair, counting re-entries. The dataset contains 224,598 pairs and 78,275 recording IDs. Themes can overlap. A linear model of $\log(1 + \text{chart days})$ includes indicators for romance, explicitness and conspicuous consumption together, plus market, first observed chart year, broad genre and catalogue release-group effects. Standard errors group all appearances of a recording together across markets. Songs already charting at the start or still charting at the end have incomplete lifetimes; the outcome is observed chart presence, not a survival estimate or a causal effect of lyrical content.

4. Results

4.1 Thematic profile and market differences

Figure 2 reports all 26 dimensions. Romantic affection (35.5%) and heartbreak (21.4%) are the most frequent clearly present themes in the full selected corpus. Money references (16.5%), reciprocity (14.1%), drugs/alcohol (13.7%), and materialism (13.5%) follow. Sexual explicitness is 7.2%, objectification 5.8%, and misogyny 3.4%. Themes overlap: their percentages do not sum to 100%.

Brazil's all-history repertoire contains 7,186 classified IDs out of 9,619 charted IDs (74.7% coverage). Romantic

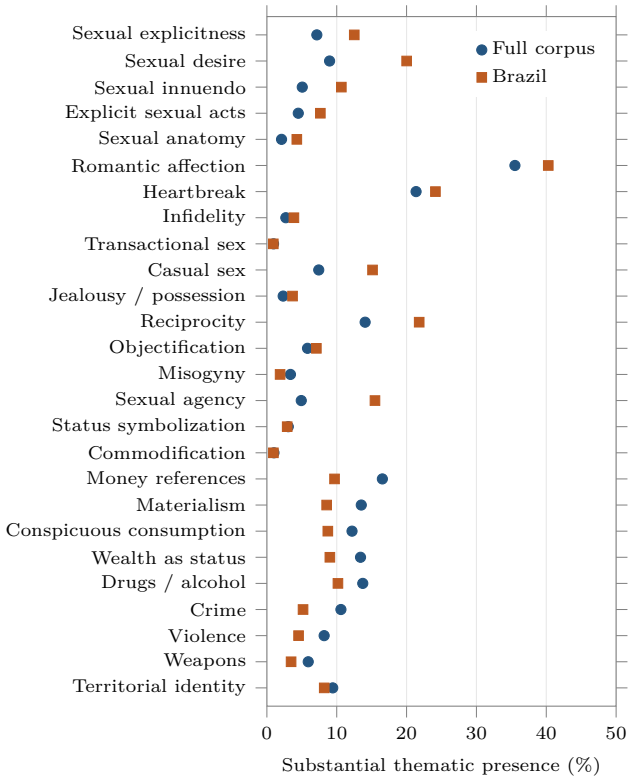


Figure 2: All taxonomy dimensions. The full corpus counts 154,964 selected recording identities once; Brazil counts 7,186 classified chart IDs once across the historical window. The full corpus is a pooled corpus reference, not an average national prevalence.

Table 2: Selected circulation markets, combined historical period. n: classified IDs; Cov.: retained share of initial repertoire; E: explicitness; C: casual sex; M: misogyny; Mat.: materialism. All thematic columns are percentages at intensity ≥ 2.

Market	n	Cov.	E	C	M	Mat.
BR	7,186	74.7	12.5	15.1	1.9	8.6
US	11,296	75.6	16.1	15.2	8.3	24.5
GB	10,752	74.5	10.9	10.9	3.8	17.9
DE	11,907	73.5	9.4	8.9	7.2	27.7
FR	11,579	75.9	10.3	8.7	6.8	23.0
ES	8,082	73.4	11.4	13.6	1.6	10.3
PT	8,065	73.4	11.9	12.6	3.8	14.2
JP	8,047	63.5	3.6	3.7	0.8	4.9

affection is 40.3%, heartbreak 24.1%, casual sex 15.1%, explicitness 12.5%, objectification 7.1%, misogyny 1.9%, and materialism 8.6%. Brazil therefore combines comparatively frequent sexual and romantic content with lower misogyny and materialism than several comparator repertoires (Table 2). Sexual content and misogyny do not move as one measure.

Across all 68 markets, the largest explicitness shares are in the Dominican Republic (21.4%), Panama (19.2%), and Chile (17.6%). The largest misogyny share is in Romania (11.1%); the largest materialism share is in Switzerland (27.8%). These are distributions of charted recordings under a common coding rule, not rankings of national character.

Themes are distinct but overlap. Of 11,082 recordings with substantial explicitness, 60.1% also have substantial objectification and 30.8% misogyny. Among 20,943 materialism-positive recordings, 40.7% also code crime; among 12,704 violence-positive recordings, 78.3% also code crime. These conditional shares concern texts, not people. Consent, representation and stance remain separate measures in the accompanying distributions.

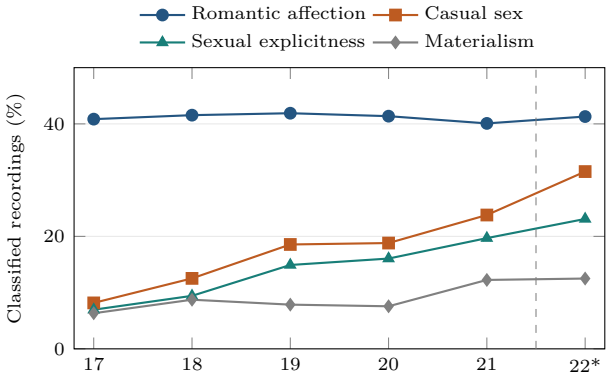


Figure 3: Brazil's annual thematic shares. Classified denominators for 2017–2022: 2,228; 1,740; 1,234; 1,414; 1,412; 368. *2022 ends on 13 March. Lines connect annual summaries, not individual song trajectories.

4.2 Brazil over time and repertoire persistence

From 2017 to 2021, Brazil's explicitness share rises from 7.0% to 19.7% and casual-sex share from 8.2% to 23.8%; romantic affection stays near 40–42% (Figure 3). Classification coverage in those years ranges from 72.6% to 76.1%. These are changes in each year's repertoire; repeated recordings may enter several annual denominators. The 2022 partial window is shown separately and is not a full-year continuation.

In the separate MGD Top-200 dynamics dataset, Brazil has 5,010 distinct recording–market pairs, with a median of 16 chart days (interquartile range 2–98), compared with 6 in the United States and United Kingdom, 7 in Germany, and 13 in Japan. Brazil's mean Top-50 Jaccard turnover is 0.069 across 1,892 adjacent daily chart pairs, against 0.114 in the United States and 0.032 in Japan. This supports a description of different repertoire persistence, without treating turnover as psychological anxiety.

Genre profiles are examined alongside exposure below. Artist rankings and network files remain supplementary descriptions rather than additional independent song samples.

4.3 International shifts and new patterns

Status and conflict gain repertoire share. In the fixed 55-market panel, mean substantial money references increase by 4.50 percentage points (pp), drugs/alcohol by 3.80, materialism by 3.78, crime by 3.71, weapons by 3.53 and territorial identity by 3.12 from 2017 to 2021 (Figures 4–5). Growth is geographically broad: weapons increase in 53/55 markets, crime in 51/55, territorial identity in 50/55, and materialism in 48/55. This is a shift in chart repertoire composition, not a claim that listeners changed their conduct.

Romance persists while heartbreak grows. Romantic affection averages 40.67% in 2017 and 40.04% in 2021; heartbreak rises from 22.18% to 25.11% in 44 markets. Sexual explicitness rises from 7.67% to 11.25%, whereas misogyny changes only from 2.71% to 2.95%. These trajectories separate affection, suffering, sexual content and gender hostility rather than treating them

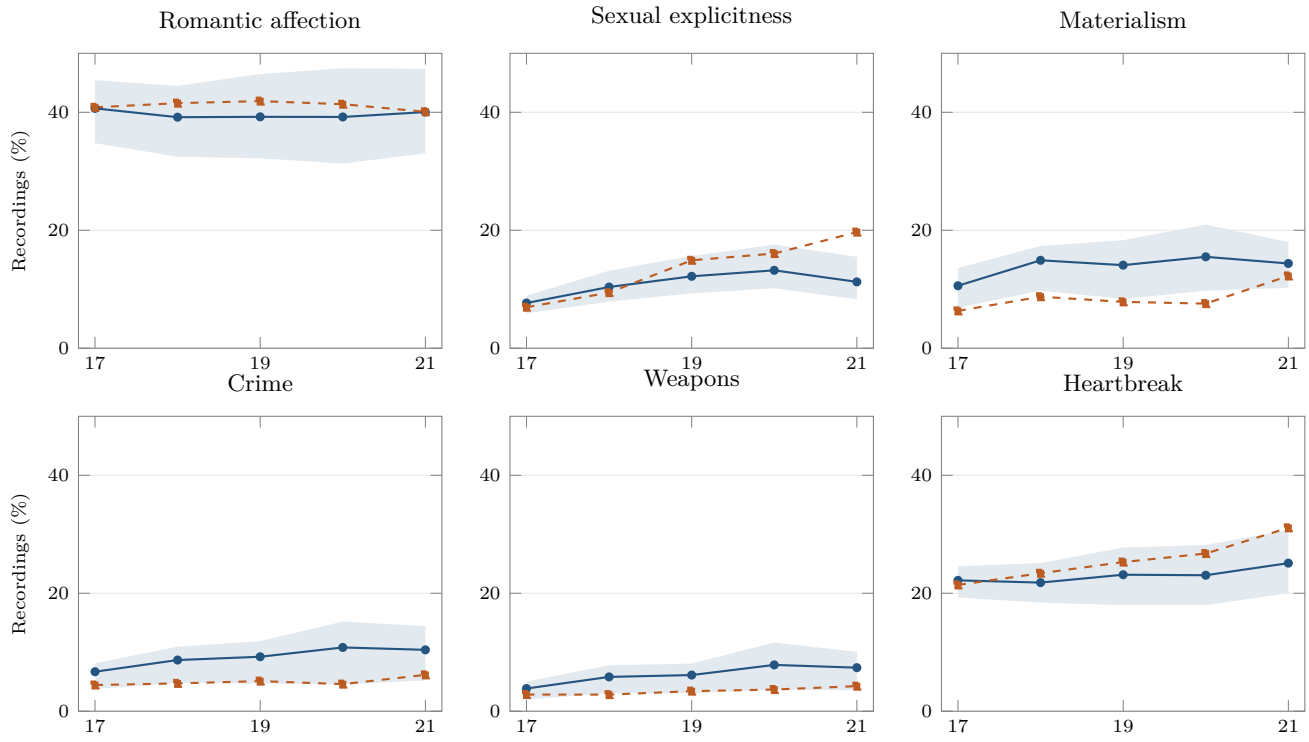


Figure 4: International trajectories, 2017–2021. Blue circles: equal-market means for the same 55 markets in all five years; orange dashed squares: Brazil. Bands are the middle 50% of market values, not confidence intervals. Each theme uses the same 0–50% axis; substantial presence means intensity ≥ 2. Markets have at least 300 classified recordings and 50% classification coverage in every year. The changing annual repertoires, not changes within a song, are measured.

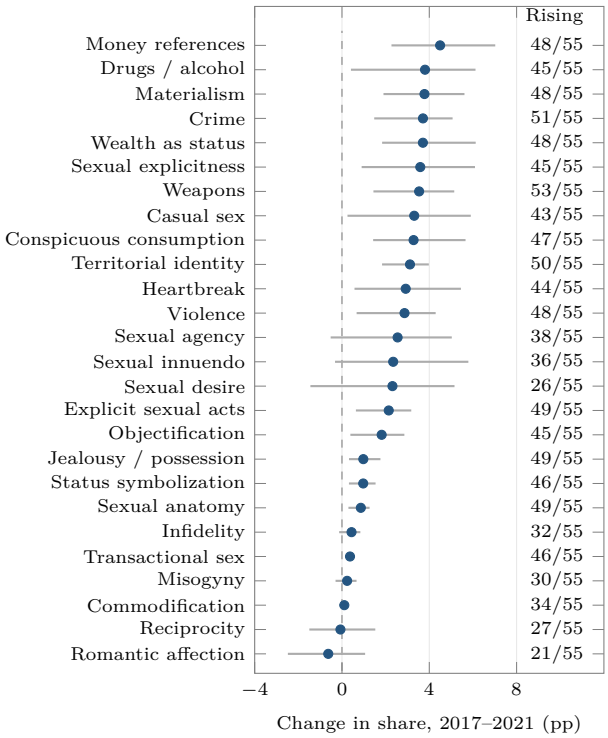


Figure 5: Changes in all 26 dimensions in the fixed 55-market panel. Blue points: mean paired market change; gray segments: interquartile range of market changes. Right-hand counts show markets with a positive change. "pp" means percentage points. These spreads describe observed markets, not sampling confidence.

as one process.

Mean growth can be geographically concentrated. Sexual desire has a positive mean change (+2.31 pp), yet its median market change is −0.09 pp and only

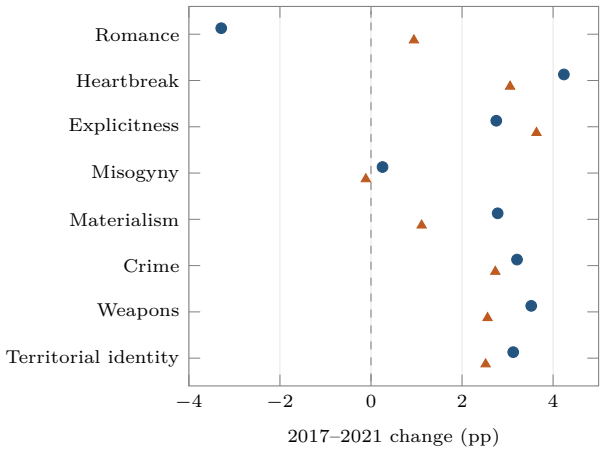


Figure 6: Chart-family contrast in the same 52 markets. Blue circles: Top 200 (source union, IDs deduplicated); orange triangles: Viral 50. Each panel has at least 300 classified IDs and 50% classification coverage in every full year. The romance trend changes direction across chart families; growth of status and conflict themes persists.

26/55 markets increase. Large shifts in a subset of markets raise the mean. Figure 5 separates average magnitude from geographical breadth.

Brazil's strongest shifts differ from the international pattern. Brazil has the largest casual-sex increase in the 55-market panel (+15.63 pp), the second-largest explicitness increase (+12.73 pp, after Andorra), and the third-largest heartbreak increase (+9.68 pp, after Paraguay and Bolivia). Its weapons increase (+1.42 pp) is smaller than the international mean (+3.53 pp). Brazil's change therefore concentrates on sexual and relational themes rather than reproducing the strongest conflict trend.

Table 3: Brazilian genre-tag profiles, MGD Top 200, 2017–2021. The eight largest artist-derived tags are selected by charted recording count. n: classified IDs; Cov.: classification coverage; R: romance; E: explicitness; C: casual sex; M: misogyny; Mat.: materialism. Theme values are percentages of classified IDs, not genre-wide population estimates. Tags overlap.

Tag	n	Cov.	R	E	C	M	Mat.
Pop	757	72.0	51.7	5.3	7.8	0.3	4.1
Agronejo	283	58.1	61.1	1.1	9.2	0.0	0.7
Arrocha	227	71.6	55.9	7.0	13.7	0.4	0.9
Rap / hip-hop	106	65.0	16.0	25.5	21.7	3.8	40.6
Brazilian funk	141	89.8	25.5	22.0	29.1	0.7	5.7
Sertanejo	84	73.0	70.2	0.0	8.3	0.0	1.2
Rock	71	63.4	31.0	0.0	0.0	0.0	0.0
K-pop	55	58.5	34.5	0.0	0.0	0.0	9.1

The aggregate masks chart-family divergence. For the same 52 markets, romance falls by 3.29 pp in Top 200 but rises by 0.94 pp in Viral 50 (Figure 6). Heartbreak grows in both (+4.24 and +3.06 pp), as do materialism (+2.78 and +1.11) and weapons (+3.52 and +2.56). Near-flat aggregate romance is therefore not evidence of an unchanged chart landscape. Misogyny also differs slightly in direction (+0.25 versus −0.11 pp).

Growth of materialism, crime, weapons, heartbreak and territorial identity persists in every source/coverage sensitivity. The distinct-recording pool also increases in these themes: materialism rises from 8.34% to 15.88%, crime from 7.03% to 13.21%, and weapons from 3.77% to 7.71%. Pool and market-mean levels answer different questions; shared recordings are not summed as independent global songs. Source-specific declines in romance remain explicit rather than being hidden by a single pooled statistic.

4.4 What receives chart exposure, and where themes concentrate

Counting recordings and counting plays give different pictures. In 2021, romance covers 40.9% of classified recording IDs in the 51-market MGD panel but 48.3% of classified chart streams. Heartbreak rises from 26.8% to 33.3%, and explicitness from 11.5% to 15.7%. Crime falls from 10.2% to 7.2%, weapons from 7.3% to 4.3%, and materialism from 15.6% to 13.2% when streams supply the weights (Figure 7). Growth in the number of conflict-themed recordings therefore coexists with a smaller share of recorded plays than a song-count measure suggests. Romance remains central to both views.

Brazil's sexual and relational themes receive extra chart weight. Among its classified Top-200 repertoire in 2021, casual sex covers 22.7% of recording IDs and 30.0% of chart streams; explicitness covers 17.3% and 23.7%; heartbreak covers 35.3% and 39.7%. Misogyny stays close to 2.3–2.4%. These are shares of classified records and their observed chart plays; the remaining unclassified streams are reported through coverage, not assigned a theme.

Genres reveal different combinations, not one national profile. Across 2017–2021, the eight largest Brazilian tag groups have distinct profiles. Sertanejo has 70.2% romantic affection and no substantial explicitness among 84 classified IDs. Brazilian funk has 29.1% casual sex and 22.0% explicitness among 141 IDs, while misogyny is 0.7%. Rap/hip-hop has 40.6% materialism and 25.5% explicitness among 106 IDs; agronejo has 61.1% romance and 1.1% explicitness among 283. Table 3 reports every group and its coverage. Country totals combine styles with different profiles. Separating these observed groups makes the comparison more informative. The tags describe artists' genre metadata and can overlap.

Table 4: Exact-year descriptive Spearman associations. n counts markets; no causal or significance claim is assigned to these correlations. Income is per-capita PPP income; GII and SIGI are distinct gender indices.

Lyrical theme / context	Year	n	ρ
Objectification / GII	2021	65	+0.158
Misogyny / GII	2021	65	-0.330
Objectification / SIGI	2019	54	-0.072
Misogyny / SIGI	2019	54	-0.459
Materialism / Gini	2021	48	-0.456
Materialism / income (PPP)	2021	67	+0.408
Materialism / youth unemployment	2021	66	+0.316
Violence / total homicide	2021	60	-0.261
Crime / total homicide	2021	60	-0.253
Violence / female homicide	2021	59	-0.234
Violence / partner-family deaths	2021	39	-0.124
Romance / youth unemployment	2021	66	-0.267
Heartbreak / youth unemployment	2021	66	-0.119
Reciprocity / GII	2021	65	+0.440
Sexual agency / GII	2021	65	+0.360
Wealth as status / income	2021	67	+0.410
Territorial identity / urbanization	2021	67	+0.183
Explicitness / internet access	2021	67	-0.227
Explicitness / OONI anomalies	2021	41	-0.039

Social links keep their direction, but change strength with exposure. With the same complete-data markets in each comparison, the 2021 materialism–income correlation drops from 0.525 with recording weights to 0.319 with stream weights (50 markets). Materialism–Gini weakens from −0.430 to −0.229 (36); misogyny–GII from −0.430 to −0.232 (50); violence–homicide from −0.329 to −0.181 (44). Each direction persists under all three weightings in all five years. These links are stronger in repertoire composition than in the chart-stream view in 2021. Measures of musical variety and recorded exposure should therefore be reported separately when connecting music to social conditions.

4.5 Music and six social axes

Within the 386 historical market–year cells, World Bank PPP income and internet use are observed in 380 cells each, GII in 372, Gini in 286, and total UNODC homicide in 332. SIGI is observed for 54 markets in 2019. These differences determine the pairwise sample sizes in Table 4.

All 26 lyric dimensions are compared with 13 downloaded contextual indicators, separately in 2017–2021: 1,690 specified cells, of which 1,586 support descriptive correlations. The matrix includes observed and insufficient-data cells, without selecting only large coefficients. Table 4 organizes selected contrasts across the six axes; the complete matrix accompanies the article.

Materialism is positively associated with income ($\rho = 0.408$) and negatively with Gini (−0.456) in 2021. Misogyny is negatively associated with GII in 2021 (−0.330) and SIGI in 2019 (−0.459), whereas objectification has weaker associations with both. Violence and crime shares are negatively associated with total homicide (−0.261 and −0.253). The direction and magnitude differ by theme and context; lyrical content is not a direct proxy for national violence or gender conditions.

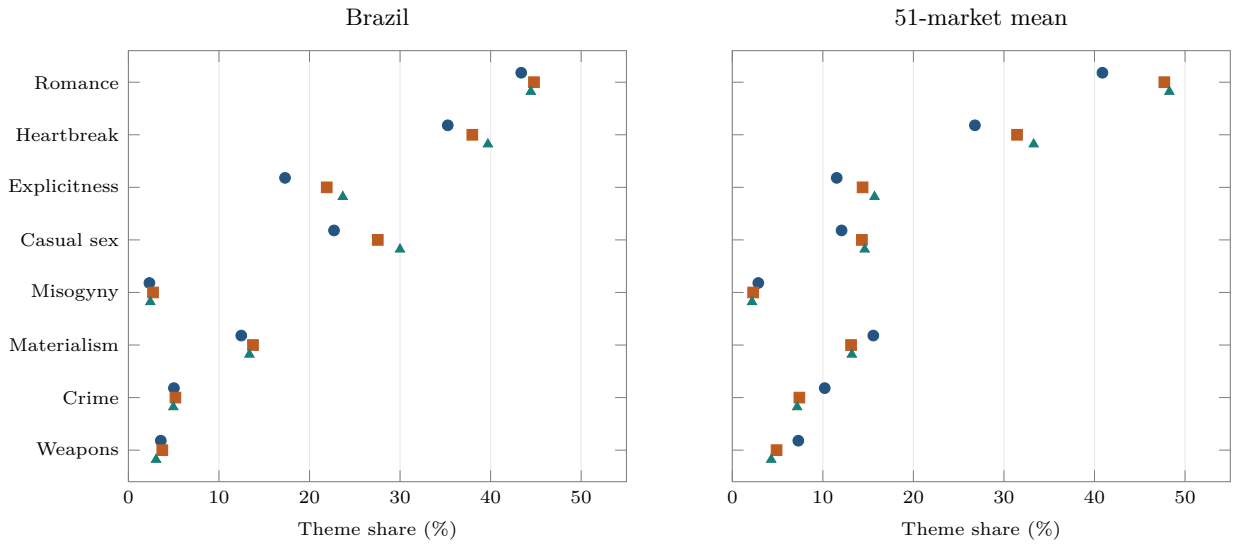


Figure 7: Three views of the same classified Top-200 repertoire in 2021. Blue circles count recording IDs equally; orange squares weight their observed chart days; teal triangles weight observed chart streams. The right panel averages the same 51 markets equally. Every denominator contains classified records only. Brazil has 1,035 classified IDs out of 1,382 charted IDs, covering 71.9% of chart days and 69.4% of chart streams. Median stream coverage across the 51 markets is 73.1% (range 45.5–80.9%). Streams are chart-eligible plays during chart appearances, not people or all Spotify plays.

Intimacy connects romantic affection and heartbreak to youth unemployment. **Gender and power** connects reciprocity and sexual agency to GII, with patterns distinct from misogyny. **Status** connects wealth symbolism to income. **Digital and cultural environments** connect explicitness to internet access and territorial identity to urbanization. Table 4 reports every coefficient and sample size for these contrasts. OONI anomaly share has a near-zero association with explicitness. These are separate questions about representation, social conditions, infrastructure and circulation.

Brazil's 2021 contextual profile anchors the comparison: Gini 52.9, youth unemployment 28.3%, GII 0.405, internet use 80.7%, urban population 87.1%, and total homicide 21.7 per 100,000. These actual downloaded values are joined to the same-year musical repertoire; they are not inferred from lyrics.

Social links persist across years, not just in a single cross-section. Holding the market sample fixed within each pair, materialism correlates positively with PPP income in all five years (ρ = 0.41–0.61, 54 markets) and negatively with Gini (−0.50–−0.30, 39 markets). Misogyny and GII remain negatively associated (−0.55–−0.43, 53 markets); violence and homicide also remain negative (−0.43–−0.27, 47 markets). Territorial identity and urbanization remain positive (0.23–0.36, 54 markets). Romance's link to youth unemployment is weaker (−0.17–−0.04, 53 markets). Figure 8 shows these fixed-sample trajectories. The results challenge a direct reading of lyrical violence or misogyny as national social prevalence; they instead define concrete questions about repertoire selection, circulation and representation.

4.6 Four questions about music and society

Cultural Evolution: How did references to wealth, weapons, and romantic affection evolve across different world regions between 2017 and 2021? Wealth-as-status references rise from 8.44% to 11.79% in the Latin American panel and from 11.02% to 14.37% in the Nordic panel, but fall from 5.49% to 3.39% in Japan and Taiwan. Weapons rise by 1.86 percentage points in Latin America and 5.04 in Nordic markets, while their change in Japan and Taiwan is close to zero. Romantic affection stays near 46.3% in Latin America, falls from 35.91% to 34.10% in Nordic markets and rises from 47.04% to 51.69% in Japan and Taiwan. Table 5 reports changes and recording-level intervals; the Nordic romance interval includes zero. These are changes in circulating lyrics, not direct measurements of residents' values.

Table 5: Change from 2017 to 2021, percentage points, with 95% recording-bootstrap intervals. Wealth means wealth as status; romance means romantic affection. East Asia contains Japan and Taiwan only.

Panel	Wealth	Weapons	Romance
Latin America	+3.34 [+1.73, +4.97]	+1.86 [+1.03, +2.85]	+0.04 [-2.85, +2.65]
Nordic countries	+3.35 [+1.98, +4.64]	+5.04 [+3.85, +6.24]	-1.81 [-3.70, +0.18]
East Asia	-2.10 [-3.18, -1.01]	-0.11 [-0.75, +0.52]	+4.65 [+2.13, +7.40]
Southeast Asia	-0.82 [-2.37, +0.73]	+0.88 [+0.01, +1.64]	+3.72 [+1.02, +6.47]
Anglophone markets	+4.10 [+2.35, +5.61]	+7.36 [+5.98, +8.68]	-1.58 [-3.48, +0.43]
Other panel markets	+5.59 [+4.09, +6.89]	+4.34 [+3.25, +5.35]	-2.13 [-3.82, -0.44]

The decomposition helps explain why these shifts cannot automatically be called cultural change. Of the 3.78-point materialism increase, the genre partition, including untagged songs, assigns 2.57 points to changes within groups (95% interval 1.22–3.73) and 1.21 to changing group shares (0.77–1.66). For weapons, the corresponding terms are 2.87 (2.15–3.50) and 0.66 (0.43–0.93); for crime they are 2.82 (1.98–3.77) and 0.89 (0.55–1.23). Changes within a group still include different songs entering and leaving its chart repertoire; the words of the same recording have not been observed changing.

Missing genre tags are central to this result: untagged songs account for 2.54 of the 2.57-point within-group

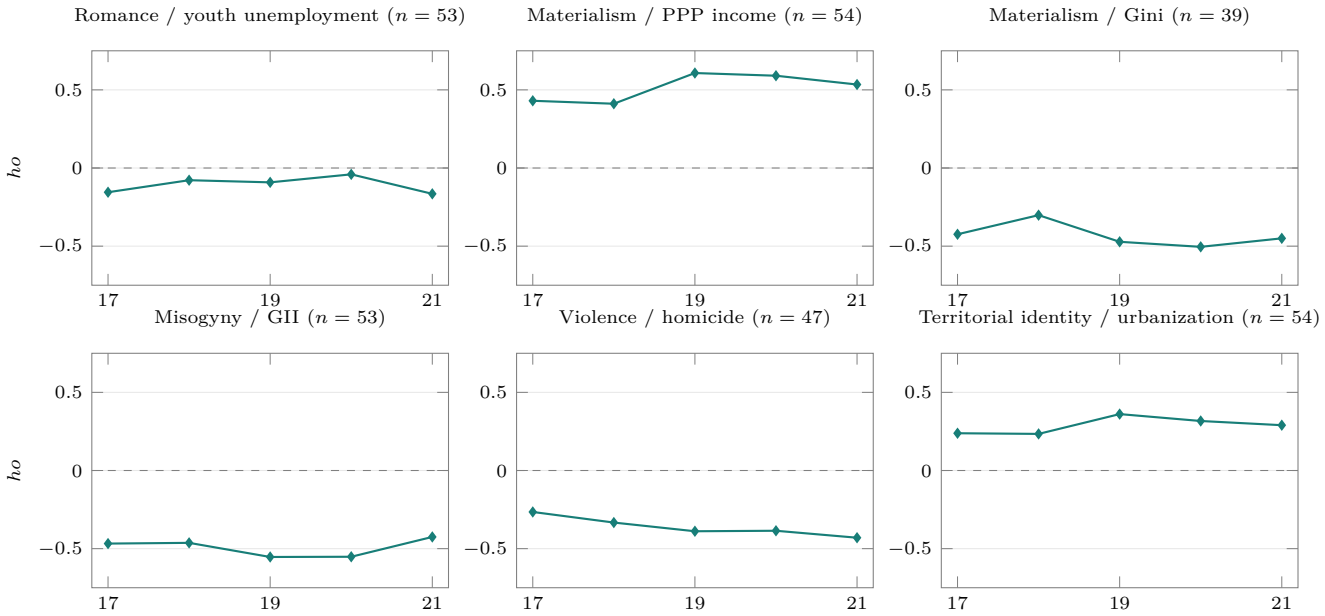


Figure 8: Persistence of social associations, using a fixed market sample within each pair in every year. Points are separate-year Spearman correlations; n counts markets, not people. World Bank data supply income, Gini, youth unemployment and urbanization; UNDP supplies GII; UNODC supplies homicide. No interpolation or pooled-year correlation is used. Stable directions describe aggregate associations and do not establish causes.

materialism component. The tagged groups together account for only 0.03 points. Meanwhile, the mean rap/hip-hop group share rises from 10.72% to 17.37%, contributing +2.49 points to composition before offsets from other groups. The aggregate within-group term therefore cannot establish that materialism increased within known genres. Better metadata are needed to separate that possibility from changes within the untagged repertoire.

The separate release-group decomposition assigns materialism −1.90 points within groups (−3.04 to −0.64) and +5.68 to composition (4.46–6.85). For weapons the terms are +0.68 (−0.01 to 1.41) and +2.85 (2.10–3.59); for crime, −0.08 (−0.87 to 0.77) and +3.79 (2.93–4.75). Thus, newer catalogue groups gaining chart share provide a plausible compositional account of much of the aggregate increase. Missing dates, reissues and genre tags limit the interpretation; a language decomposition remains pending.

Music & Society: Do national income inequality (Gini) or homicide rates predict the prevalence of materialism or violence in local top-chart music? The available analysis establishes associations, not validated prediction. Across the same 39 complete-data markets, annual materialism–Gini correlations are negative, ranging from −0.50 to −0.30 in 2017–2021. Across 47 markets, violence–homicide correlations range from −0.43 to −0.27. These patterns do not support a simple claim that more unequal or more violent countries have more materialistic or violent chart lyrics. In the separate weighted Top-200 panel, the 2021 correlations weaken from −0.430 to −0.229 for materialism–Gini and from −0.329 to −0.181 for violence–homicide when streams replace recording counts. No held-out predictive accuracy has been estimated. Chart selection, shared international hits and social confounding remain possible explanations.

Clustering & Regionality: Can we identify cultural clusters of music consumption based on lyric themes (e.g., Latin America vs. Nordic countries vs. East Asia)? We identify four exploratory clusters of chart-repertoire themes, with mean silhouette 0.442. All 16 Latin American panel markets fall in one cluster, alongside Nicaragua and Spain. Japan, Taiwan and South Korea share another cluster with India and several Southeast Asian markets, as well as Andorra and Turkey. Nordic markets are less uniform: Denmark, Finland, Norway and Sweden share a cluster, while Iceland belongs to another. The remaining clusters contain a mix of European, Anglophone and other markets. The grouping therefore partly follows geography and cultural circulation, but does not reproduce a simple continental map. It describes thematic similarity in charted songs, not a tested classification of individual listening cultures. All memberships are exported; cluster stability has not been established.

Streaming Longevity: Do songs featuring high romantic affection stay on charts longer than songs high in sexual explicitness or material conspicuous consumption? Romance-positive recordings have a median of 11 observed chart days, compared with 7 when romance is below the threshold. Explicitness-positive recordings have a median of 14 days, compared with 7 otherwise; conspicuous-consumption-positive recordings have 8 days, the same median as their comparison group. These overlapping groups cannot be interpreted as mutually exclusive song types.

In the joint adjusted model, romance and explicitness both associate with longer chart presence. Their log(1 + days) coefficients are 0.231 (95% interval 0.177–0.285) and 0.325 (0.249–0.401); conspicuous consumption is 0.050 (−0.012 to 0.112). The romance-minus-explicitness contrast is −0.095 (−0.184 to −0.005), whereas romance minus conspicuous consumption is +0.180 (0.105–0.256). These exploratory comparisons

Table 6: Adjusted within-market search associations. Effect units: percentage points per local search-index standard deviation. All models have 59 markets and 290 market–years; intervals are 95%; q covers the 30-model family. Full baseline and 70% sensitivity estimates accompany the manuscript.

Outcome	β	95% interval	q
Sexual explicitness	-0.491	[-1.016, +0.035]	0.285
Objectification	-0.197	[-0.532, +0.137]	0.563
Misogyny	+0.004	[-0.187, +0.196]	0.965
Materialism	-0.653	[-1.105, -0.201]	0.078
Infidelity	-0.099	[-0.265, +0.066]	0.563
Casual sex	-0.431	[-1.014, +0.152]	0.432
Transactional sex	-0.023	[-0.083, +0.036]	0.722
Drugs / alcohol	-0.578	[-1.019, -0.138]	0.083
Violence	-0.069	[-0.463, +0.326]	0.862
Crime	-0.168	[-0.618, +0.282]	0.722

support longer observed presence for romance than for conspicuous consumption after adjustment, but not longer presence than for explicitness. They concern chart days within 2017–2021, not lifetime popularity, all-platform listening or a causal advantage from adding a theme to a song.

4.7 Historical search associations

The established search specification uses the English term *porn*, selecting the lexicographically first query batch within each market before fitting outcomes. Local indices are standardized within each eligible market series. Models use 2017–2021, at least 300 classified recordings per market–year, at least 80% temporal search coverage, at least four eligible years per market, and at least 20 markets. Native-language terms remain in the descriptive panel.

For ten focal themes, linear models compare changes within each market while accounting for market differences and shared year effects. Adjusted models also account for PPP income and internet use. Effects express a change in musical share, in percentage points, for one local standard deviation of search interest. Uncertainty is grouped by market. The baseline includes 60 markets and 295 market–years; the higher-coverage check includes 24 and 104; adjusted models include 59 and 290. Benjamini–Hochberg correction covers all 30 models.

No model meets $q < 0.05$. Adjusted materialism is -0.653 points per local standard deviation (95% interval $[-1.105, -0.201]$, $q = 0.078$), and drugs/alcohol is -0.578 ($[-1.019, -0.138]$, $q = 0.083$). Explicitness is -0.491 ($[-1.016, 0.035]$, $q = 0.285$). The 70% sensitivity has wider intervals and does not retain evidence below the same threshold. The findings do not support a simple positive relationship between relative adult-term searches and sexual lyrical content. They also do not prove that no relationship exists.

5. Interpretation and scope

Music provides a measurable record of how intimacy, power, status, violence and identity circulate. Status and conflict themes expand across repertoires, while romance retains greater chart-stream weight. Brazil combines rising sexual themes, strong relational content and distinct genre profiles. The chart-family split, the exposure measures and the genre profiles each explain a different part of this picture. Social correlations remain visible over time but weaken under some exposure weights. A single country score would hide these differences.

The atlas makes these findings actionable for research and public debate: educators can compare representations, researchers can test mechanisms, and cultural institutions can track repertoire change with explicit evidence. It replaces broad judgments with inspectable measures. Each international position is specific to one dimension; no composite moral ranking is imposed.

Inference is bounded by the observed units. Charted songs are a selected repertoire, not a sample of residents; streams are not unique listeners or endorsement. Spotify chart streams use eligibility filtering [2]. Equal recording weights describe repertoire diversity, not listening exposure. Top 200 and Viral 50 have different selection mechanisms. Shared songs create dependence between markets. The new recording-level bootstrap and longevity model retain that dependence; the earlier search models still group uncertainty only by market. Short panels, changing repertoires, incomplete genre and language metadata, missing contextual values and unmeasured classification error constrain interpretation.

Adult-term search interest is not pornography consumption. OONI coverage comprises 211 market–year cells with connectivity-test counts; the tests come from participating networks and anomalies do not by themselves confirm censorship [8]. Female homicide is not synonymous with femicide, and partner/family homicide is a distinct indicator. The 2019 SIGI edition and annual GII must retain their own dates and definitions. Annual 2022 context extends beyond the musical cutoff and stays outside models.

Historical commercial visits, weighted WVS sexual-value aggregates and dated TikTok exposure are absent. Contemporary DNS and the traffic pilot remain separate. A legacy index combining historical song shares with 2024 population is synthetic, not an exposed audience estimate. Viral-50 membership does not identify TikTok effects on longevity.

6. Reproducibility and conclusion

The analysis exports all 26 dimensions, country/year denominators, eight international panels, chart-day and stream weights, genre profiles, fixed-sample social links, the complete contextual matrix and the 30 search models. The revision adds regional changes, two repertoire decompositions, exploratory cluster memberships, longevity summaries and adjusted contrasts, alongside the blind human-review sheets and validation protocol. The manifests record the database snapshot, model versions, hashes, the reported specialist check and the pending independent validation. Publication figures are also supplied as PDF, SVG and PNG.

The supplied scripts reproduce the frozen statistics, new comparisons, contextual matrix and standalone LaTeX figures without new model calls or database writes. The contribution is a traceable comparison of lyrical themes, circulation and social context across markets and years. Independent human validation and language metadata are the main remaining requirements for stronger claims and submission readiness.

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International atlas: all 68 circulation markets

Position ranks sexual explicitness (E) from highest to lowest. Other columns retain their own values; the companion interactive atlas ranks any of the 26 dimensions. n is the classified recording count; Cov. is the retained share of the initial chart repertoire; R is romantic affection; C is casual sex; O is objectification; M is misogyny; Mat. is materialism; V is violence; Cr. is crime. All thematic values are percentages at intensity ≥ 2, pooled over 2017–2021 and through 13 March 2022. Each recording counts once per market. Higher position is a larger thematic share, not better performance. Brazil is the reference case.

Pos.	Circulation market	n	Cov.	R	E	C	O	M	Mat.	V	Cr.
1	Dominican Republic (DO)	5,723	72.8	40.5	21.4	23.0	13.8	4.1	19.4	9.0	10.2
2	Panama (PA)	5,159	71.8	43.9	19.2	21.3	10.9	2.8	15.3	6.7	7.8
3	Chile (CL)	6,311	72.0	42.8	17.6	20.3	10.1	2.2	12.0	5.7	6.4
4	Paraguay (PY)	4,842	71.6	43.6	17.2	20.7	9.6	1.7	11.2	4.6	5.3
5	Romania (RO)	5,867	75.2	30.5	16.8	15.9	14.9	11.1	24.3	14.0	16.3
6	Honduras (HN)	4,427	70.9	45.4	16.5	18.7	9.1	2.1	12.8	5.3	6.0
7	United States (US)	11,296	75.6	31.8	16.1	15.2	12.7	8.3	24.5	16.0	17.5
8	Nicaragua (NI)	3,345	64.2	47.6	15.5	17.5	8.5	1.8	11.0	5.1	5.4
9	Canada (CA)	10,296	74.7	32.8	15.3	14.7	12.3	7.7	24.3	15.0	16.4
10	El Salvador (SV)	4,421	70.8	47.0	15.2	18.1	8.2	1.9	10.9	4.5	5.2
11	Uruguay (UY)	5,725	72.0	42.6	14.9	18.3	8.7	1.7	11.2	4.6	5.4
12	Bolivia (BO)	4,506	70.4	49.4	14.5	17.9	7.6	1.3	9.2	3.3	3.8
13	Bulgaria (BG)	5,078	70.2	30.4	14.4	14.2	12.2	7.3	23.6	14.7	15.3
14	Ecuador (EC)	5,057	70.7	48.3	14.2	17.4	7.4	1.2	9.4	3.4	3.7
15	Costa Rica (CR)	5,708	72.0	42.2	14.1	16.0	8.6	2.8	13.1	7.0	7.1
16	Guatemala (GT)	5,282	72.3	47.3	14.0	17.1	7.6	1.7	10.1	4.2	4.8
17	Iceland (IS)	6,123	64.2	29.7	13.9	13.3	11.8	7.3	22.8	14.4	15.9
18	South Africa (ZA)	6,293	71.7	33.2	13.4	12.5	10.5	6.1	23.0	12.8	13.4
19	Colombia (CO)	6,422	70.4	46.1	13.1	16.2	6.9	1.4	9.0	4.0	4.4
20	Argentina (AR)	6,525	72.4	41.9	13.1	15.3	8.2	1.9	10.9	4.8	5.5
21	Peru (PE)	5,202	70.3	48.9	12.9	16.4	6.8	1.3	8.2	3.1	3.5
22	**Brazil (BR)**	**7,186**	**74.7**	**40.3**	**12.5**	**15.1**	**7.1**	**1.9**	**8.6**	**4.5**	**5.2**
23	Greece (GR)	6,788	71.4	31.4	12.5	12.0	10.8	7.1	24.2	15.4	18.6
24	United Arab Emirates (AE)	3,123	71.6	42.7	12.4	11.9	9.4	5.5	18.3	12.6	12.6
25	Portugal (PT)	8,065	73.4	35.3	11.9	12.6	7.8	3.8	14.2	8.6	9.2
26	Switzerland (CH)	13,028	73.3	28.2	11.5	11.2	10.4	7.3	27.8	16.2	22.0
27	Spain (ES)	8,082	73.4	42.1	11.4	13.6	6.2	1.6	10.3	5.0	6.0
28	Luxembourg (LU)	5,135	70.3	34.0	11.3	11.3	9.3	5.3	21.7	12.7	15.9
29	Ukraine (UA)	2,707	72.9	25.9	11.2	10.2	10.6	7.1	18.0	10.8	11.5
30	Israel (IL)	4,547	63.9	36.2	11.0	11.0	8.7	4.9	15.5	10.1	9.9
31	Latvia (LV)	6,303	70.0	31.4	11.0	10.7	9.6	6.1	18.4	11.2	12.0
32	United Kingdom (GB)	10,752	74.5	33.1	10.9	10.9	7.9	3.8	17.9	13.3	15.0
33	Morocco (MA)	3,433	68.3	25.0	10.7	9.4	9.2	6.8	23.9	16.5	21.8
34	New Zealand (NZ)	7,950	70.9	35.4	10.6	9.9	7.5	4.4	14.6	10.1	10.2
35	Russia (RU)	3,286	74.9	23.7	10.4	9.2	9.6	6.2	16.3	11.1	11.3
36	Slovakia (SK)	7,133	69.8	29.1	10.4	10.4	8.9	5.5	18.8	9.5	10.8
37	France (FR)	11,579	75.9	23.8	10.3	8.7	8.9	6.8	23.0	19.7	27.6
38	Belgium (BE)	10,156	71.3	30.8	10.1	9.4	8.5	5.5	20.8	13.4	17.7
39	Netherlands (NL)	10,718	68.5	30.6	10.0	10.4	8.7	4.6	24.5	10.2	15.4
40	Estonia (EE)	5,750	69.8	35.0	10.0	9.4	7.7	4.5	15.0	9.2	9.4
41	Australia (AU)	7,991	71.8	35.8	9.8	9.5	6.8	3.8	14.0	9.8	9.8
42	Saudi Arabia (SA)	2,894	62.1	42.9	9.6	9.4	7.5	4.1	14.0	9.7	9.5
43	Denmark (DK)	9,255	72.7	32.4	9.4	9.7	7.1	3.7	17.6	9.5	12.0
44	Germany (DE)	11,907	73.5	26.0	9.4	8.9	9.1	7.2	27.7	15.2	21.1
45	Mexico (MX)	5,640	70.5	47.6	9.4	11.9	5.2	1.1	7.9	4.1	4.8
46	Hungary (HU)	7,943	71.0	34.5	9.1	9.4	7.0	4.1	13.3	8.5	8.5
47	Ireland (IE)	9,608	72.9	36.4	9.0	9.0	6.3	3.3	12.8	10.2	10.3
48	Austria (AT)	10,850	73.7	29.6	9.0	8.8	8.5	6.2	25.0	13.0	17.5
49	Lithuania (LT)	6,176	69.0	36.4	9.0	8.7	7.1	4.2	14.2	9.1	9.4
50	Czechia (CZ)	8,589	69.6	29.9	8.9	8.5	7.3	5.0	15.1	8.3	9.0
51	Norway (NO)	10,109	69.5	35.9	8.7	9.1	6.4	3.5	12.6	7.7	8.5
52	Italy (IT)	9,394	75.5	34.0	7.8	7.5	6.2	3.9	14.9	8.0	10.8
53	Sweden (SE)	10,844	72.0	36.5	7.2	7.3	4.7	2.2	12.9	11.1	12.6
54	Egypt (EG)	2,334	66.8	37.2	6.9	6.4	5.3	3.2	14.1	11.1	8.3
55	Andorra (AD)	5,235	74.6	44.5	6.8	8.6	4.2	1.4	6.2	4.5	4.2
56	Finland (FI)	10,913	68.4	32.5	6.6	7.2	4.8	2.8	9.7	8.1	7.0
57	Poland (PL)	8,908	70.0	30.0	6.5	6.9	5.5	3.7	13.3	6.5	7.7
58	Singapore (SG)	6,848	69.0	50.2	5.5	5.5	3.2	1.6	7.7	3.9	3.6
59	India (IN)	3,402	63.0	51.4	5.4	5.6	3.9	2.1	8.8	6.2	5.8
60	Malaysia (MY)	6,624	69.6	51.5	5.1	5.1	3.1	1.6	7.6	3.8	3.7
61	Philippines (PH)	6,273	71.8	55.7	4.8	4.6	2.4	1.1	5.5	2.9	2.7
62	Türkiye (TR)	6,350	66.0	37.3	4.5	4.3	3.5	1.9	8.9	6.9	6.7
63	Vietnam (VN)	4,220	64.9	57.2	4.1	4.4	2.4	0.9	5.7	2.4	2.4
64	Thailand (TH)	4,923	62.6	59.9	4.1	4.3	2.5	1.1	5.7	2.4	2.4
65	South Korea (KR)	1,969	79.1	52.8	3.8	4.0	2.1	1.0	8.8	2.9	2.7
66	Indonesia (ID)	5,985	70.3	56.1	3.6	4.0	1.9	0.9	4.8	2.3	2.2
67	Japan (JP)	8,047	63.5	43.5	3.6	3.7	1.9	0.8	4.9	3.0	2.7
68	Taiwan (TW)	8,887	59.7	52.4	3.5	3.9	1.8	0.8	5.2	2.5	2.3